

Experiment No. 7

Aim : Implement any one of the Local Search Techniques.

Theory :

Definition of Stimulated Annealing Algorithm –

Simulated Annealing is an optimization algorithm designed to search for an optimal or near-optimal solution in a large solution space. The name and concept are derived from the process of annealing in metallurgy, where a material is heated and then slowly cooled to remove defects and achieve a stable crystalline structure. In Simulated Annealing, the "heat" corresponds to the degree of randomness in the search process, which decreases over time (cooling schedule) to refine the solution. The method is widely used in combinatorial optimization, where problems often have numerous local optima that standard techniques like gradient descent might get stuck in. Simulated Annealing excels in escaping these local minima by introducing controlled randomness in its search, allowing for a more thorough exploration of the solution space.

Advantages of Simulated Annealing

- Simple Implementation
- Ability to Escape Local Minima
- Global Optimization
- Flexibility

Limitations of Simulated Annealing

- Parameter Sensitivity
- Computational Time
- Slow Convergence

Algorithm for implementing Simulated Annealing :

- S1] Start with the initial random solution.
- S2] Evaluate its cost.
- S3] Generate neighbor solution.
- S4] If neighbor better -> accept.
- S5] If neighbor is worst -> accept with probability.
 $P = -\Delta E/T$
 ΔE = Increase in cost
 T = Temperature
- S6] As temperature get down choosing of worst case scenario is also less.
- S7] Stop.

Code :

```
main.py    Share  Run
```

```
1 import math, random
2 def f(x):
3     return x**2 # square of x
4 x = random.uniform(-10, 10)
5 T = 1000
6 while T > 1:
7     new_x = x + random.uniform(-1, 1)
8     if f(new_x) < f(x) or random.random() < math.exp(-(f(new_x) - f(x)) /
9         T):
10         x = new_x
11     T *= 0.95
12 print("Best x :", x, "f(x) :", f(x))
```

Output :

```
Output Clear  
Best x : -0.007087883969076003 f(x) : 5.02380991590846e-05  
=== Code Execution Successful ===
```

Conclusion :

In this experiment, we implemented the Simulated Annealing algorithm and observed its ability to escape local minima and move toward near-optimal solutions. The results show that, with proper parameter tuning, it is a simple yet effective method for solving complex optimization problems.